

General Topology 1

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1 Problems

Q 1. Let Λ be an uncountable set. For each $\lambda \in \Lambda$ associate a real $r_\lambda > 0$. Is $\sum_{\lambda \in \Lambda} r_\lambda$ infinite?

Proof: r_Λ is bounded, otherwise we can pick r_λ to be arbitrarily large. Let M be an upper bound. Define $A_n = (\frac{1}{n}, M]$. Each $r_\lambda > 0$, so there exists n such that $r_\lambda \in A_n$. Hence, $r_\Lambda = \cup_{n=1}^{\infty} A_n$. Moreover, in each A_n there are a finite number of r_λ . r_Λ is a countable union of finite sets, so r_Λ is also countable (or finite), contradiction.

E 2.7: Any open subset of $\mathbb{R}$ is a countable union of open intervals.

Let $A \subset \mathbb{R}$ be an open set. Define the equivalence relation $\sim$ on A such that $x \sim y$ if there exists an open interval $I = (a, b)$ for which $\{x, y\} \subset I \subset A$. $\sim$ is reflexive since A is open. $\sim$ is symmetric. $\sim$ is transitive. We have

$$\{x, z\} \subset \{x, y\} \cup \{y, z\} \subset I \cup I',$$

where I' is an interval containing $\{y, z\}$. Since the open intervals meet at y , their union is another open interval. Consider the equivalence classes of $\sim$. Each class C is an interval since if $x, y \in C$, $x < y$, there exists an open interval I such that $\{x, y\} \subset I$, so $(x, y) \subset I$. Moreover, C is open since $x \sim x$. Hence, A is a union of disjoint open intervals.

A has at most two unbounded interval components. Consider the remaining elements A' of A . Each (nonempty) interval has positive size. Since the sum of uncountably many positive reals is infinite, there are countably many remaining intervals in A' , so A is a union of countably many intervals.

Better (rigorous) finish: each interval contains a distinct rational.

2A.1: $\rho(f, g)$ is well-defined because of EVT on $|f - g|$. $\rho(f, g)$ is nonnegative. $\rho(f, g) = 0$ means for all $x \in I$, $|f(x) - g(x)| \leq 0$, or $f(x) = g(x)$ for all $x \in I$. Hence $f \equiv g$. $\rho(x, y) = \rho(y, x)$ because of the absolute value.

$\rho(f, g) + \rho(g, h) \geq |f(x) - g(x)| + |g(x) - h(x)| \geq |f(x) - h(x)|$ for all $x \in I$, so $\rho(f, g) + \rho(g, h) \geq \sup_{x \in I} |f(x) - h(x)| = \rho(f, h)$.

2A.2: σ is well-defined. σ is nonnegative. The integral of a continuous non-negative function f equals 0 when $f \equiv 0$. Integrating the triangle inequality gives the triangle inequality for σ .

2A.3: We can pick $f(x_0) = g(x_0)$ but $f \neq g$.

2C.1: $\varphi(x, x) = 0$ by assumption. φ is symmetric. If $\varphi(x, y) = 0$ and $\varphi(y, z) = 0$, $\varphi(x, z) \leq 0$, so $\varphi(x, z) = 0$.

2C.2: Let (M, ρ) be a pseudometric. Define $\rho^*([x], [y]) = \varphi(x, y)$. ρ^* is well defined: suppose $x \sim x'$ and $y \sim y'$. $\rho(x, y) \leq \rho(x, y') + \rho(y', y) = \rho(x, y')$. Similarly, $\rho(x, y') \leq \rho(x, y)$, so $\rho(x, y) = \rho(x, y')$. $\rho(x', y) = \rho(x', x) + \rho(x, y) \geq \rho(x, y)$. Similarly, $\rho(x, y) \geq \rho(x', y)$, so $\rho(x, y) = \rho(x', y)$. $\rho(x, y) = \rho(x, y') = \rho(x', y')$, so ρ^* is well-defined.

ρ^* is nonnegative. $\rho^*([x], [y]) = 0$ means $\rho(x, y) = 0$, or $x \sim y$, or $[x] = [y]$. If $[x] = [y]$, then $\rho^*([x], [y]) = 0$. ρ^* is symmetric.

$$\begin{aligned} \rho^*([x], [y]) + \rho^*([y], [z]) &= \rho(x, y) + \rho(y, z) \\ &\geq \rho(x, z) \\ &= \rho^*([x], [z]). \end{aligned}$$

2C.3: If $U(x, \epsilon) \subset A$, then $U([x], \epsilon) \subset h(A)$ for all $x \in A$. For all $y \in A$ such that $\rho(y, x) < \epsilon$, we have $y \in A$. Let $[y] \in M^*$ such that $\rho^*([y], [x]) < \epsilon$, so $\rho(y, x) < \epsilon$, or $y \in A$. Then $[y] \in h(A)$.

Let $B \subset M^*$ be open and A be such that $h(A) = B$. Let $x \in A$. There exists ϵ such that $U([x], \epsilon) \subset B$, or that for all $y \in M$ such that $\rho(x, y) < \epsilon$, $y \in A$.
2C.5: ?

Let ρ and φ be metrics on M . Then $(M, \rho + \varphi)$ is a metric space.

Proof: $\rho(x, y) + \varphi(x, y) \geq 0$. $\rho(x, y) + \varphi(x, y) = 0$ iff $x = y$. $\rho(x, y) + \varphi(x, y) = \rho(y, x) + \varphi(y, x)$. $\rho(x, y) + \varphi(x, y) + \rho(y, z) + \varphi(y, z) \geq \rho(x, z) + \varphi(x, z)$.

Definition of continuity: for all discs V centered at $f(x)$, there exists a disc U centered at x such that $f(U) \subset V$.

Generalize: for all open sets V containing $f(x)$, there exists an open set U containing x such that $f(U) \subset V$.

$\text{id} : (M, \rho) \rightarrow (M, \rho)$ is continuous. If $V \subset M$ is any open set, take $U = V$, so that $\text{id}(U) = V$.

Let $f : (M, \rho) \rightarrow (N, \varphi)$ be continuous. Let $A \subset f(M)$ be open. Then $f^{-1}(A)$ is open. Try: take $f : I \rightarrow \mathbb{R}$ be continuous. $f(I)$ is a closed interval by IVT. Pick $x \in f^{-1}(A)$. $f(x) \in A$, so there exists an open neighborhood V containing A . Then there exists open $U \subset M$ containing x such that $f(U) \subset V \subset A$. Hence, $U \subset f^{-1}(A)$.

If $f : (M_1, \rho_1) \rightarrow (M_2, \rho_2)$ and $g : (M_2, \rho_2) \rightarrow (M_3, \rho_3)$ are continuous, then $g \circ f : (M_1, \rho_1) \rightarrow (M_3, \rho_3)$ is also continuous.

For any open $V \subset M_3$ containing $g(f(x))$, there exists open $U_1 \subset M_2$ containing $f(x)$ such that $g(U_1) \subset V$. There exists open $U_2 \subset M_1$ containing x such that $f(U_2) \subset U_1$. Hence, $g(f(U_2)) \subset g(U_1) \subset V$.

Suppose we have the stronger $\rho(x, y) + \rho(y, z) \geq \sqrt{2}\rho(x, z)$. Then ρ^2 is also a metric.

Proof: nonnegativity, zero iff, symmetry. Triangle:

$$\rho(x, y)^2 + \rho(y, z)^2 \geq \frac{1}{2}(\rho(x, y) + \rho(y, z))^2 \geq \rho^2(x, z).$$

Suppose $f : (M, \rho) \rightarrow (N, \varphi)$ such that $\rho(x, y) = \varphi(f(x), f(y))$. f is injective: if $f(x) = f(y)$, $\rho(x, y) = 0$, so $x = y$.

Suppose further f is surjective. Let $U(f(x), \epsilon) \subset N$ and let $f(y)$ lie in this disc. Then $\rho(x, y) = \varphi(f(x), f(y)) < \epsilon$, so $y \in U(x, \epsilon)$. I have shown $f^{-1}(U(f(x), \epsilon)) \subset U(x, \epsilon)$, so f^{-1} is continuous.

$U(f(x), \epsilon) \subset f(U(x, \epsilon))$??? Equal with bijective continuous map?

Let $A \subset M$. Show that $U(A, \epsilon)$ is indeed open. Proof: let $x \in U(A, \epsilon)$, $r = \epsilon - \rho(A, x) > 0$. Consider $y \in U(x, r)$. $\rho(y, A) = \inf \rho(y, a) \leq \rho(y, x) + \rho(x, a) < \epsilon - \rho(A, x) + \rho(x, a)$ for all $a \in A$. Hence, $\rho(y, A) < \epsilon - \rho(A, x) + \inf \rho(x, a) = \epsilon$. That is, $U(x, r) \subset U(A, \epsilon)$.

Bounded metrics: a metric ρ is said to be bounded if $\rho(x, y) \leq A$ for some A for all $x, y \in M$. Claim: $\rho^*(x, y) = \min(\rho(x, y), 1)$ is also a metric.

Nonnegative, zero iff, symmetric. Suppose $\varphi(x, z) \leq 1$. Only check $\varphi(x, z) \leq \varphi(x, y) + \varphi(y, z)$ (true). Else $\varphi(x, z) \geq 1$. Only check $1 \leq \varphi(x, y) + \varphi(y, z)$ (true). Create bounded metric out of any metric.

Let $f(x) \in M^*$ and $\epsilon > 0$. If $\epsilon > 1$, $U(f(x), \epsilon) = M^*$. $x \in M$ and M is open. $f(M) \subset M^*$, AWD. Else $\epsilon \leq 1$. By continuity there exists $f(U(x, \delta)) \subset$

$U(f(x), \epsilon)$. Pick $\delta' = \min(\delta, 1) \dots$? A disc in M^* is either a disc in M or M itself. Suppose f is continuous in M . Consider a disc in M^* . If the disc is M , we're done as we pick $f(M) \subset M$. Else, by continuity, there is a disc U in M such that $f(U) \subset V$. If the radius of U is greater than 1, pick a smaller disc U' so that $f(U') \subset f(U) \subset V$, so that U' is a disc in M^* .

Isometrics. I is isometric to any closed interval of $\mathbb{R}$ of the same length. Suppose $J = [a, a + 1]$. Pick $f : I \rightarrow J$ by $f(x) = x + a$.

Show $\mathbb{R}$ is not isometric to $\mathbb{R}^2$. There are infinitely many points on the unit circle in $\mathbb{R}^2$. Suppose $x_0 = f^{-1}(0, 0)$. Consider $f^{-1}(\mathcal{U})$, which has infinitely many points, each of which is 1 away from x_0 by the isometry, contradiction (there are only two points in $\mathbb{R}$ one away from x_0).

Is $\mathbb{R}^2$ isometric to $\mathbb{R}^3$? Consider a circular intersection of two unit spheres centered at P, Q in $\mathbb{R}^3$. $P' = f^{-1}(P)$, $Q' = f^{-1}(Q)$ are distinct, so the number of points in $\mathbb{R}^2$ having distance 1 from both P', Q' is finite, contradiction.

Guess: $\mathbb{R}^n \cong \mathbb{R}^m$ iff $n = m$.

Note: unbounded is isometric to unbounded, bounded is isometric to bounded.

Are there examples where $M \cong N$, but M, N look different? Or is isometry like isomorphism?

Let $\mathcal{P} \subset \mathbb{R}^3$ lie on a plane. Suppose $\mathcal{Q} \cong \mathcal{P}$. Then does $\mathcal{Q}$ lie on a plane?

Open set minus closed set is open. $A - B = A \cap B^C$. A, B^C open, so $A - B$ is open. Closed minus open is closed? $(A - B)^C = (A \cap B^C)^C = A^C \cup B$ is open, so $A - B$ is closed.

Prove: $\max(d_A(B), d_B(A)) + \max(d_B(C), d_C(B)) \geq \max(d_A(C), d_C(A))$. $d(a, b) + d(b, c) \geq d(a, c)$. $d(a, B) + d(B, c) \geq d(a, c)$. $d_B(A) + d_B(C) \geq d(a, c) \geq d(A, c)$. $d_B(A) + d_B(C) \geq d_A(C) \dots$

A sequence (x_n) is said to converge to a point x if for all open discs V centered at x , there exists $N \geq 1$ such that $n \geq N$, $x_n \in V$. Given that (x_n) converges, x is unique.

Show that a set $A \subset M$ is closed iff all convergent sequences $(x_n) \subset A$ converge to a point in A . Suppose FTSOC (x_n) converges to a point not in A . A^C is open, so there exists $U(x, \epsilon) \subset A^C$. By limit, pick N such that $x_N \in U(x, \epsilon)$. $x_N \in A^C$, contradiction.

Show: $d_A(B) + d_B(A) + d_B(C) + d_C(B) \geq 2d(a, b) + 2d(b, c)$ for all a, b, c .

Isometric metric spaces. Let $(M, \rho), (N, \varphi)$ be metric spaces. Suppose $f : M \rightarrow N$ is a surjective function such that $\rho(x, y) = \varphi(f(x), f(y))$. Then f is surjective and both f, f^{-1} are continuous.

Proof: Let $U(f(x), \epsilon) \subset N$. Let $y \in U(x, \epsilon)$. Then $\rho(x, y) = \varphi(f(x), f(y))$, so $f(y) \in U(f(x), \epsilon)$. That is, $f(U(x, \epsilon)) \subset U(f(x), \epsilon)$. Suppose $f(x) = f(y)$. Then $\rho(x, y) = \varphi(f(x), f(y)) = 0$, so $x = y$. f is bijective. Replacing x, y with $f^{-1}(a), f^{-1}(b)$ we have the same initial conditions, so f^{-1} is also continuous.

For all $a \in A$, $d(a, B) \leq d(A, B) < \epsilon$, so $a \in U(B, \epsilon)$, so $A \subset U(B, \epsilon)$. Similarly, $B \subset U(A, \epsilon)$.

Exists. Nonnegative. Zero iff. Symmmetric. Triangle. None.

A function $f : X \rightarrow Y$ is continuous iff for all closed $B \subset Y$, $f^{-1}(B)$ is closed. Suppose B is closed. Then B^C is open, so $f^{-1}(B^C)$ is open. Then $f^{-1}(B^C)^C = f^{-1}(B)$ is closed.

Let $[x] \in h(A)$. There exists $U(x, \epsilon) \subset A$. Consider $[y] \in U([x], \epsilon) \subset M^*$, so $d(y, x) < \epsilon$, or $y \in U(x, \epsilon) \subset A$. Hence $[y] \in h(A)$. That is, $U([x], \epsilon) \subset h(A)$, so $h(A)$ is open if A is open.

Let $B \subset M^*$ be open and A be such that $h(A) = B$. Let $x \in A$. $h(x) \in B$. There exists $U([x], \epsilon) \subset B$. Consider $y \in U(x, \epsilon)$. $d(y, x) = d([y], [x]) < \epsilon$.